MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 2**

Limits: Chapter 1, Sections 1.5–1.7 (the limit of a function; calculating limits; the precise definition).

Key facts to have at hand

- $\lim_{x\to a} f(x) = L$ means: $f(x)$ can be made **as close to** L **as we like** by taking x close enough to a, $x \neq a$. The value $f(a)$ — defined or not — plays no role: a limit is about the *approach*, not the *arrival*.
- **One-sided limits** $\lim_{x\to a^-} f(x)$, $\lim_{x\to a^+} f(x)$ use only $x < a$, resp. $x > a$. $\lim_{x\to a} f(x) = L$ *if and only if* both one-sided limits exist and equal L.
- **Infinite limits:** $\lim_{x\to a} f(x) = \infty$ means $f(x)$ becomes as large as we like for x close enough to a — a sentence about behaviour, not a number. Then $x = a$ is a **vertical asymptote**. Check the sign on each side separately.
- **Limit laws** (all limits at $x \to a$ assumed to exist): $\lim[f \pm g] = \lim f \pm \lim g$; $\lim[cf] = c \lim f$; $\lim[fg] = \lim f \cdot \lim g$; $\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ *provided* $\lim g \neq 0$; $\lim[f]^n = [\lim f]^n$; $\lim \sqrt[n]{f} = \sqrt[n]{\lim f}$ (for even n: $\lim f > 0$).
- **Direct substitution:** if f is a polynomial or a rational function and a is in its domain, $\lim_{x\to a} f(x) = f(a)$.
- $0/0$ **is an instruction, not an answer:** rewrite, then substitute — factor and cancel; multiply by the conjugate; combine fractions; go one-sided when the formula changes at a (absolute values, cases). Two functions that agree for all $x \neq a$ have the same limit at a.
- **Squeeze Theorem:** if $g(x) \leq f(x) \leq h(x)$ near a (except possibly at a) and $\lim g = \lim h = L$, then $\lim f = L$.
- **Precise definition:** $\lim_{x\to a} f(x) = L$ means: for every $\varepsilon > 0$ there is a $\delta > 0$ such that if $0 < |x-a| < \delta$ then $|f(x) - L| < \varepsilon$. (ε is the challenge, δ the response; δ may depend on ε; a smaller δ always works too.) $\lim_{x\to a} f(x) = \infty$: for every M there is $\delta > 0$ such that if $0 < |x - a| < \delta$ then $f(x) > M$.

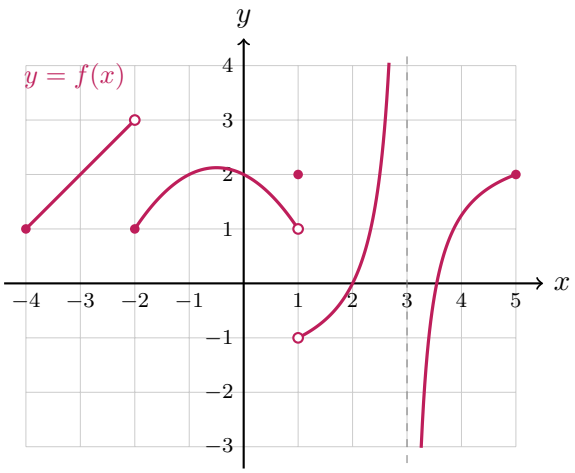
Part I · Reading and calculating limits (Sections 1.5–1.6) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If $\lim_{x\to a} f(x)$ exists, then $f(a)$ is defined.
2. If $f(a)$ is defined, then $\lim_{x\to a} f(x)$ exists.
3. If $\lim_{x\to a} f(x) = L$ and $f(a)$ is defined, then $f(a) = L$.
4. $\lim\limits_{x\to 2} \dfrac{x^2 - 4}{x - 2} = 4.$
5. If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, then $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ does not exist.
6. If $\lim_{x\to a} f(x) = 2$ and $\lim_{x\to a} g(x) = 0$, then $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ is not a real number.
7. $\lim\limits_{x\to 0} \dfrac{|x|}{x} = 1.$
8. If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$, then $\lim_{x\to a} f(x) = L$.
9. If $\lim_{x\to 0} f(x) = \infty$, then $f(0)$ is not defined.
10. $\lim\limits_{x\to 0} x \sin \dfrac{1}{x} = 0.$

Problem 1 *Reading limits off a graph.*

The graph of a function f is shown; the dashed line is a vertical asymptote.



Find each of the following. Name the value of the function. Explain.

(a) $\lim_{x\to -2^-} f(x)$ (b) $\lim_{x\to -2^+} f(x)$ (c) $\lim_{x\to -2} f(x)$

(d) $\lim_{x\to 0} f(x)$

(e) $\lim_{x\to 1^-} f(x)$ (f) $\lim_{x\to 1^+} f(x)$ (g) $\lim_{x\to 1} f(x)$

(h) $\lim_{x\to 3^-} f(x)$ (i) $\lim_{x\to 3^+} f(x)$ (j) $\lim_{x\to 3} f(x)$

Problem 2 *Using the limit laws.*

Suppose that $\lim_{x\to 2} f(x) = 3$, $\lim_{x\to 2} g(x) = -2$ and $\lim_{x\to 2} h(x) = 0$. Evaluate the following limits as $x \to 2$, naming the law you use at each step — or explain why the laws do not give the answer.

(a) $\big[f(x) + 3g(x)\big]$

(b) $f(x)\,g(x) - h(x)$

(c) $\dfrac{f(x)}{g(x)}$

(d) $\dfrac{g(x)}{h(x)}$

(e) $\sqrt{f(x)+1}\,\big[g(x)\big]^2$

(f) $\dfrac{h(x)}{f(x)}$

Problem 3 $0/0$*: factor and cancel.*

In each limit, substitution gives $0/0$. Factor, cancel the common factor — and say in one sentence why you may cancel — then substitute.

(a) $\lim_{x\to 4} \dfrac{x^2-16}{x^2-5x+4}$

(b) $\lim_{x\to -2} \dfrac{x^2+5x+6}{x^2-4}$

(c) $\lim_{x\to 3} \dfrac{2x^2-5x-3}{x-3}$

(d) $\lim_{h\to 0} \dfrac{(1+2h)^2-1}{h}$

(e) $\lim_{t\to 1} \dfrac{t^3-1}{t^2-1}$

Problem 4 $0/0$*: multiply by the conjugate.*

Again $0/0$ every time. Multiply top and bottom by the conjugate of the expression with the root, simplify, cancel, substitute.

(a) $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9}$

(b) $\lim_{h\to 0} \dfrac{\sqrt{4+h}-2}{h}$

(c) $\lim_{x\to 2} \dfrac{\sqrt{5x-1}-3}{x-2}$

(d) $\lim_{x\to 1} \dfrac{x-1}{\sqrt{x+3}-2}$ (e) $\lim_{x\to 0} \dfrac{\sqrt{2+x}-\sqrt{2-x}}{x}$

Problem 5 $0/0$ *(and $\infty-\infty$): combine the fractions first.*

Bring the fractions to one denominator, simplify, cancel, substitute. In (d) and (e) substitution gives not $0/0$ but "$\infty-\infty$", which decides nothing either.

(a) $\lim_{x\to 5} \dfrac{\frac{1}{x}-\frac{1}{5}}{x-5}$ (c) $\lim_{x\to -3} \dfrac{\frac{1}{3}+\frac{1}{x}}{x+3}$ (e) $\lim_{x\to 2}\Big(\dfrac{1}{x-2}-\dfrac{4}{x^2-4}\Big)$

(b) $\lim_{h\to 0} \dfrac{\frac{1}{4+h}-\frac{1}{4}}{h}$ (d) $\lim_{t\to 0}\Big(\dfrac{1}{t}-\dfrac{2}{t^2+2t}\Big)$

Problem 6 *One-sided limits and functions defined in pieces.*

(a) Let

$$f(x)=\begin{cases} x^2+1, & x<1,\\ 3-x, & 1\le x<3,\\ \sqrt{x-3}, & x\ge 3. \end{cases}$$

Find the one-sided limits of f at 1 and at 3; decide whether $\lim_{x\to 1} f(x)$ and $\lim_{x\to 3} f(x)$ exist, and compare them with $f(1)$ and $f(3)$. Sketch the graph.

(b) Find $\lim_{x\to 2^-} \dfrac{|x-2|}{x-2}$, $\lim_{x\to 2^+} \dfrac{|x-2|}{x-2}$ and decide whether $\lim_{x\to 2} \dfrac{|x-2|}{x-2}$ exists.

(c) Find the one-sided limits at 3 and decide whether the two-sided limit exists, for (i) $\dfrac{x^2-9}{|x-3|}$ and (ii) $\dfrac{x^2-6x+9}{|x-3|}$. Why does one of the limits exist and the other not?

(d) Find $\lim_{x\to 0}\Big(\dfrac{1}{|x|}-\dfrac{1}{x}\Big)$. (Careful: treat the two sides separately.)

Problem 7 *Infinite limits and vertical asymptotes.*

Determine these limits. Find all vertical asymptotes of these functions.

(a) $\lim_{x\to 3} \dfrac{x+2}{x-3}$ (c) $\lim_{x\to 2} \dfrac{x+2}{x^2-4}$

(b) $\lim_{x\to 0} \dfrac{x-2}{x^2(x+3)}$ (d) $\lim_{x\to 1} \dfrac{x^2-4}{x^2-3x+2}$

Problem 8 *The Squeeze Theorem.*

(a) Show that $\lim_{x\to 0} x^2\cos\dfrac{1}{x^2}=0$.

(b) If $3x-2\le f(x)\le x^2-x+2$ for all $x\ge 0$, find $\lim_{x\to 2} f(x)$.

(c) Show that $\lim_{x\to 0^+} \sqrt{x}\left(1+\sin^2\dfrac{2\pi}{x}\right)=0$.

(d) Why does the Squeeze Theorem *not* show that $\lim_{x\to 0}\cos\dfrac{1}{x}$ exists, although $-1\le\cos\frac{1}{x}\le 1$?

Part II · The precise definition of a limit (Section 1.7) · 50 minutes

Warm-up II · True or false?

Throughout, $\lim_{x\to a} f(x) = L$ is meant in the precise sense: for every $\varepsilon > 0$ there is $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$.

1. If $|f(x) - L| < 0.01$ whenever $0 < |x - a| < 0.5$, then $\lim_{x\to a} f(x) = L$.
2. In the definition, δ is allowed to depend on ε.
3. If some δ works for a given ε, then every smaller δ works for that ε too.
4. If some δ works for a given ε, then the same δ works for every smaller ε.
5. For $\lim_{x\to 3}(2x - 1) = 5$ and $\varepsilon = 0.1$, the choice $\delta = 0.05$ works.
6. For the same limit and $\varepsilon = 0.1$, the choice $\delta = 0.1$ also works.
7. The definition requires $f(a) = L$.
8. "$\lim_{x\to a} f(x) = \infty$" means that $f(a) = \infty$.

Problem 9 *Linear ε–δ proofs.*

Prove each statement from the definition. Write the scratch work (finding δ backwards) on the left and the proof (forward) on the right, separated by a line.

(a) $\lim_{x\to 2}(5x - 3) = 7$

(b) $\lim_{x\to -1}(2 - 3x) = 5$

(c) $\lim_{x\to 4}\left(\frac{x}{2} + 1\right) = 3$

(d) $\lim_{x\to -2}(4x + 9) = 1$

(e) $\lim_{x\to 1}(7 - 2x) = 5$

(f) $\lim_{x\to 3}\dfrac{x^2 - x - 6}{x - 3} = 5$

Then, for (a): which δ does your proof give when $\varepsilon = 0.01$? And is $\delta = 0.01$ itself good enough for $\varepsilon = 0.01$?

Problem 10 *When the factor is not a constant: the* min *trick.*

(a) Prove that $\lim_{x\to 2} x^2 = 4$. (Hint: $|x^2 - 4| = |x - 2|\,|x + 2|$. First agree that $|x - 2| < 1$, and find how large $|x + 2|$ can then be.)

(b) Prove that $\lim_{x\to 3}(x^2 - 2x) = 3$.

(c) Prove that $\lim_{x\to -1}(x^2 + 3x) = -2$.

(d) Prove that $\lim_{x\to 1} x^3 = 1$. (Hint: $x^3 - 1 = (x - 1)(x^2 + x + 1)$.)

(e) Prove that $\lim_{x\to 1}\dfrac{1}{x} = 1$. (Hint: $\left|\dfrac{1}{x} - 1\right| = \dfrac{|x - 1|}{|x|}$; keep x away from 0.)

Problem 11 *Limits equal to $+\infty$, precisely.*

Recall: $\lim_{x\to a} f(x) = \infty$ means that for every $M > 0$ there is $\delta > 0$ such that $f(x) > M$ whenever $0 < |x - a| < \delta$. For $x \to a^+$ replace "$0 < |x - a| < \delta$" by "$a < x < a + \delta$" (and for $x \to a^-$ by "$a - \delta < x < a$"). Prove each statement from the definition: scratch work first, then the proof forward, starting with "Let $M > 0$".

(a) $\lim_{x\to 3}\dfrac{1}{(x - 3)^2} = \infty$. Which δ does your proof give for $M = 10{,}000$?

(b) $\lim_{x\to 2^+}\dfrac{3}{x - 2} = \infty$.

(c) $\lim_{x\to 0^+}\dfrac{1}{\sqrt{x}} = \infty$. Which δ does your proof give for $M = 100$?

(d) $\lim_{x\to -1} \dfrac{5}{|x+1|} = \infty$.

(e) $\lim_{x\to 0}\left(1 + \dfrac{1}{x^2}\right) = \infty$. (Hint: you may use $1 + \dfrac{1}{x^2} > \dfrac{1}{x^2}$ — a lower bound that is easier to push above M.)

Problem 12 *Limits equal to $-\infty$, precisely.*

Write down what $\lim_{x\to a} f(x) = -\infty$ means in the same style (for every $N < 0$ there is $\delta > 0$ such that ...), and the one-sided versions. Then prove each statement from the definition. Start every proof with "Let $N < 0$".

(a) $\lim_{x\to 0^+}\left(-\dfrac{1}{x}\right) = -\infty$.

(b) $\lim_{x\to 2} \dfrac{-3}{(x-2)^2} = -\infty$.

(c) $\lim_{x\to 1^-} \dfrac{2}{x-1} = -\infty$.

(d) $\lim_{x\to 0}\left(1 - \dfrac{1}{|x|}\right) = -\infty$.

(e) $\lim_{x\to 3^+} \dfrac{x-5}{x-3} = -\infty$. (Hint: for $3 < x < 4$ the numerator is between -2 and -1; use that to find a simpler function *above* the given one.)

Problem 13 *A proof with a hole.*

A student "proves" that $\lim_{x\to 1}(3x+2) = 5$ as follows:

> *Let $\delta > 0$. Choose $\varepsilon = 3\delta$. If $0 < |x-1| < \delta$, then $|(3x+2)-5| = 3|x-1| < 3\delta = \varepsilon$. Therefore the limit is 5.*

Every inequality in it is correct. Why is it nevertheless not a proof? Rewrite it correctly.

Challenge *(for those who finish early, or at home)*

(a) Prove from the definition that $\lim_{x\to 0} x \sin\dfrac{1}{x} = 0$. (What δ works for a given ε?)

(b) Prove from the definition that $\lim_{x\to 0} \dfrac{|x|}{x}$ does *not* exist. (Suppose it were some number L, take $\varepsilon = \frac{1}{2}$, and find two points within any δ of 0 whose values cannot both be within $\frac{1}{2}$ of L.)